

ASHISH S. JOSHI

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RESEARCH PROFILE

M.S. student in Artificial Intelligence at Boston University interested in multimodal machine learning, language and vision-language models, robust perception, and reliable AI systems. My research goal is to study how models that combine visual understanding, language reasoning, and decision-making can become more robust, interpretable, and effective in real-world environments.

EDUCATION

Boston University <i>Master of Science in Artificial Intelligence</i> Relevant focus: multimodal machine learning, computer vision, machine learning, model evaluation	Boston, MA 2025 – 2027
MIT World Peace University <i>Post Graduate Diploma in Artificial Intelligence and Machine Learning</i> <i>Bachelor of Science in Computer Science</i>	Pune, India 2022 – 2023 2019 – 2022

RESEARCH INTERESTS

Multimodal Machine Learning; Language and Vision-Language Models; Model Evaluation; Mechanistic Interpretability; Robust Perception; Generative AI; Retrieval-Augmented Generation; Reliable AI Systems.

PUBLICATIONS

Unleashing the Power of Deep Learning: A Novel Approach for Defect Detection in Solid Rocket Motor

Ashish Joshi, Kaushik Joshi, Rupali Kute, Sunil Jaiswal, Prerna Mishra
FirePhysChem, 2025. DOI: 10.1016/j.fpc.2025.06.005

Selecting RAG vs. File-First Deployment Strategies Across Open-Weight Reasoning Models for Efficient AI Systems

Siddhanth Kalyanaraman, Ravi Gupta, Kaushik Sandeep Joshi, **Ashish Sandeep Joshi**, Vidney Jadhav, Jayesh Suryawanshi
IEEE Systems and Information Engineering Design Symposium (SIEDS), 2026

SELECTED RESEARCH PROJECTS

SpatialBench-VLM: Benchmarking and Interpreting Spatial Reasoning in VLMs Feb. 2026 – May 2026

Multimodal ML, Vision-Language Models, Spatial Reasoning, Interpretability

- Built a unified evaluation framework for spatial reasoning across CV-Bench-2D, VSR, Spatial-MM, SpatialMQA, SAT, and MME-RealWorld-Lite.
- Evaluated BLIP-2, InstructBLIP, Phi-3.5-Vision, Cambrian-8B, and Qwen2.5-VL across verification, counting, containment, directional, egocentric, and action-conditioned settings.
- Analyzed cross-benchmark generalization using accuracy, dataset difficulty, relation-family performance, agreement/disagreement patterns, Spearman correlation, and Kendall rank stability.
- Used Grad-CAM, attention/entropy analysis, and CKA to relate spatial reasoning failures to visual grounding and internal model behavior.

SARF: Structure-Aware Representation Fusion for Improving CLIP Transfer Mar. 2026 – Apr. 2026

Computer Vision, Representation Learning, Multimodal Fusion, Interpretability

- Proposed a lightweight fusion framework combining RGB semantic features with edge-derived structural features for low-resource circuit diagram classification.
- Compared RGB-only, edge-only, SARF-Sum, SARF-Concat, SARF-Learned Fusion, and ensemble variants using frozen CLIP ViT-B/32 embeddings with supervised fusion heads.
- Evaluated Top-1 accuracy, Macro-F1, Top-5 accuracy, balanced accuracy, class coverage, and fix-versus-hurt behavior across 32 circuit classes.
- Used Grad-CAM, attention rollout, and topology-conditioned analysis to test whether fusion improved attention to wire paths, intersections, and connectivity-critical regions.

RARC: Adaptive Retrieval Controller for Reliable LLM Question Answering Jan. 2026 – Mar. 2026

Retrieval-Augmented Generation, LLMs, Adaptive Retrieval, Reliable AI

- Built a reasoning-aware RAG controller that dynamically adjusts retrieval depth, context width, and answer mode based on query complexity.

- Evaluated RARC against Fixed RAG and No-Retrieval baselines across 768 runs with Llama 3.3 70B, Qwen2.5 32B, multiple quantization settings, and different corpus sizes.
- Measured hallucination, groundedness, answer relevance, latency, GPU utilization, and memory to analyze reliability and deployment trade-offs.

SlideMIL-ViT: Transformer-Based Weak Supervision for Gigapixel Visual Prediction *MedAI Hackathon, 2026*
Computer Vision, Vision Transformers, Multiple Instance Learning, Weak Supervision

- Built a weakly supervised visual prediction pipeline for gigapixel whole-slide images using Multiple Instance Learning over UNI2-h vision embeddings.
- Developed transformer-enhanced MIL with attention aggregation, top- k discriminative patch selection, feature normalization, and slide-to-patient prediction modeling.
- Designed leakage-safe patient-level cross-validation and evaluated robustness with ROC-AUC, log loss, calibration analysis, and multi-fold generalization, contributing to a top-3 finish at the BU MedAI Hackathon.

Decision Twin: Accountable AI System for High-Stakes Decisions *Nov. 2025 – Dec. 2025*
LLMs, Retrieval-Augmented Generation, AI Governance, Risk Modeling

- Built a policy-grounded LLM decision system combining business simulation, risk modeling, retrieval, and self-critique for accountable recommendations.
- Modeled pricing and growth scenarios across revenue, profit, churn, CAC, retention, and expansion dynamics.
- Used Qwen2.5-14B to generate recommendations, counterarguments, failure conditions, approval requirements, and audit-ready decision traces.

Mean-Shift Image Segmentation with Color, Grayscale, and Gradient Features *May 2025 – Aug. 2025*
Computer Vision, Image Segmentation, Feature Design

- Implemented mean-shift segmentation for grayscale and color images using spatial and appearance-based feature representations.
- Extended the pipeline with gradient-based features to study how edge information affects object boundaries, texture preservation, and region consistency.
- Evaluated segmentation behavior across structured and highly textured scenes, analyzing when gradients sharpen boundaries and when texture introduces noisy regions.

TECHNICAL SKILLS

ML & Perception – PyTorch, deep learning, representation learning, weak supervision, robust visual prediction, model fine-tuning

Multimodal AI – vision-language models, LLMs, CLIP-style models, multimodal fusion, visual-language reasoning, RAG

Computer Vision – visual grounding, spatial reasoning, image segmentation, Vision Transformers, whole-slide image modeling, Grad-CAM

Intelligent Systems – agentic AI workflows, policy-grounded reasoning, decision-support systems, simulation modeling, reliability-aware pipelines

Tools – Python, C++, Linux, Git, CUDA, GPU/HPC workflows, HuggingFace, NumPy, Pandas, Matplotlib, LaTeX

RELEVANT COURSEWORK

Multimodal Machine Learning; Principles of Machine Learning; Image and Video Computing; Deep Learning; Computer Vision; Statistical Methods; Machine Learning Systems.

WORK EXPERIENCE

High Energy Materials Research Laboratory (HEMRL), DRDO *Pune, India*
Apprentice AI Engineer *Aug. 2023 – May 2025*

- Led applied AI research projects in deep learning and computer vision, working across data acquisition, dataset design, model selection, training, validation, and experimental evaluation.
- Designed controlled experimental pipelines for benchmarking neural networks on real-world scientific and industrial datasets.
- Conducted failure analysis and generalization studies to understand model behavior across different data conditions and guide iterative model improvements.

Research Intern *Feb. 2023 – Jul. 2023*

- Conducted experimental research on CNN and transformer-based vision models, studying architecture design, training behavior, representation learning, and evaluation metrics.
- Built reproducible PyTorch pipelines for hyperparameter exploration, controlled comparisons, and research documentation.

StackLab.in *Pune, India*
AI-ML Engineer Intern *Jul. 2022 – Dec. 2022*

- Worked on applied machine learning problems involving data modeling, representation learning, model evaluation, and research-oriented documentation.